Archived auxiliary benchmarks for the affine-invariant ensemble samplers

This note documents the *auxiliary* posterior benchmarks that were run alongside the headline DES Y1 3×2pt cosmology experiment but are not part of the main paper's DES-focused results. The DES Y1 experiment lives in `astro/` (see `des_section_v1_many_samplers.tex` and `des_section_v2_focus_on_pickles.tex`); the raw outputs for the benchmarks below were moved to `astro/archive/results/` to keep the working directory focused. Everything here is fully reproducible from the archived code and the package samplers.

1 Posterior suite

These seven posteriors span the geometric regimes where each sampler class is expected to win, lose, or reveal open problems. Together with DES Y1 ($D = 21$) they form the eight-problem "regime map."

problem	D	data	provenance	geometry
`pantheon`	2	1590 SNe	real (Pantheon+SH0ES)	smooth Gaussian, $\kappa \approx 4 \times 10^5$ decoupled
`hogg`	5	20 pts	synthetic / pedagogical	curved + heavy-tail mixture
`transit`	7	3000 flux	synthetic (`jaxoplanet`)	banana + sharp transit edges
`gw_pop`	8	70 ev.$\times$2k MC	synthetic	multimodal (5/8 dims)
`multi_transit`	15	5000 flux	synthetic (`jaxoplanet`)	3 sharp transit features
`multi_transit_reparam`	15	same	synthetic	Kipping + logit + log reparam
`illcond`	50–200	—	synthetic stress test	smooth Gaussian, $\kappa = e^6$

Table 1: Auxiliary benchmark posteriors. Only `pantheon` uses real astronomical observations; the rest are synthetic / pedagogical setups chosen to isolate specific geometric regimes. (`funnel` was also scaffolded but not used in the final analysis.)

Data provenance.

- `pantheon` — real Pantheon+SH0ES distance moduli and covariance (Scolnic et al. 2022, arXiv:2202.04077; data CC-BY from the public DataRelease). Fetched by `code/download_pantheon.sh`.
- `hogg` — the 20-point illustrative dataset from Hogg et al. 2010 (arXiv:1008.4686), Table 1; the mixture-with-outliers line-fitting model is ubiquitous in astronomy but these specific points are constructed.
- `transit` / `multi_transit` — the real `jaxoplanet` limb-darkened Keplerian forward model (Foreman-Mackey et al. 2024, arXiv:2402.11375) with *synthetic* light curves (fiducial truth + white Gaussian noise).

- `gw_pop` — a synthetic O3-style hierarchical catalog (Talbot–Thrane 2018 power-law-plus-peak), per-event posteriors drawn from the true population for reproducibility.
- `illcond` / `funnel` — canonical synthetic HMC stress tests (ill-conditioned Gaussian, $\kappa = e^6$; Neal's funnel).

2 Samplers

All posteriors were run through the same sampler set as DES Y1, from the `affine_invariant_samplers` package: the affine-invariant ensemble HMC family (`peaches`, `peams`, `peanuts`, `pickles`), the non-affine gradient ensemble (`chess`), `malt`, `aldi`, `gndr`, `langevin_walk`, the gradient-free ensemble moves (`stretch`, `walk`), the single-chain NUTS (`nuts`), and reference `emcee`.

3 Headline finding (ill-conditioned Gaussian scaling)

The clean dimension-scaling result is on the smooth ill-conditioned Gaussian ($\kappa = e^6 \approx 400$ across all D): per-gradient ESS is the dimension-invariant fair metric. Affine-invariant ensembles maintain efficiency from $D = 50$ to $D = 200$, whereas `emcee` fails at $D \geq 100$ because the stretch move needs walkers $\geq 2D$ and we cap at 128 walkers — a structural limitation of gradient-free affine invariance in high dimension. Per-problem CSV summaries (`results/sweep_illcond{20,50,100,200}.csv`, `sweep_multi_transit*.csv`, `sweep_summary*.csv`) hold the raw leaderboard rows.

4 How to reproduce

1. Install the package: `pip install -e .` from the repo root (brings in JAX, jax-cosmo, numpyro; `pip install emcee` for the `emcee` baseline; `pip install jaxoplanet` for the transit problems).
2. Restore the auxiliary code into `astro/`: the likelihood modules `likelihoods.py` (hogg, pantheon), `likelihoods_extra.py` (funnel, illcond), `likelihoods_gw.py`, `likelihoods_transit.py`, `likelihoods_multi_transit.py`, `likelihoods_multi_transit_reparam.py` are in `astro/archive/code/`.
3. Run the all-problems driver `astro/archive/code/run_benchmark_all_problems.py` (the untrimmed runner), e.g.

```
python -m astro.run_benchmark --problem hogg     --sampler peaches
python -m astro.run_benchmark --problem illcond  --sampler pickles --dim 100
python -m astro.run_benchmark --problem transit  --sampler emcee
```

4. For `pantheon`, first fetch the data with `code/download_pantheon.sh`.

Raw per-run outputs (`samples.npz` + `metrics.json`) for every problem × sampler × seed are under `astro/archive/results/<problem>/<sampler>/`. The `illcond_d200` run set (3.3 GB) was deleted to save disk; it is regenerable from the driver above with `--dim 200`.